SCHOOL OF COMPUTING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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10214AD701- MAJOR PROJECT INHOUSE

CLIMATE ANALYZER: DEEP LEARNING FOR VEGETATION STRESS DETECTION AND SHORT-TERM RISK FORECASTING

SEMESTER END REVIEW

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• Introduction

- ## Brief intro to the project topic

- Climate change is significantly affecting agriculture by altering rainfall patterns, temperature, and vegetation health. Early detection of vegetation stress and prediction of short-term climate risks are essential for farmers and planners to take timely decisions.
- This project, **“Climate Analyzer,”** proposes a deep learning-based framework that integrates satellite imagery and climate data to monitor vegetation health and forecast potential risks. It uses **Sentinel-2 satellite images** to analyze vegetation conditions through NDVI and applies a **Convolutional Neural Network (CNN)** to detect stressed regions. At the same time, a **Long Short-Term Memory (LSTM)** model predicts future rainfall and temperature trends.
- By combining these outputs, the system generates **risk maps (low, medium, high)** along with alerts and trend insights, enabling better decision-making in agriculture and environmental management.

- ## Motivation behind choosing the topic

- The motivation for this project comes from the growing impact of **climate change on agriculture and food security**. Farmers often face unexpected losses due to drought, temperature changes, and poor rainfall, mainly because they lack **early warning systems**.
- Existing solutions usually focus only on **either vegetation monitoring or climate forecasting**, but not both together, which limits accurate decision-making. Also, many systems require **manual data labeling or expensive resources**, making them less practical.
- To address these challenges, we chose this topic to develop a **simple, cost-effective, and automated system** that combines satellite data and climate forecasting using deep learning. The goal is to provide **early detection of vegetation stress and short-term risk prediction**, helping farmers and planners take timely actions and reduce losses.

- ### Problem Statement (1-2 lines)

- Existing systems fail to provide **integrated and timely detection of vegetation stress and climate risk**, as they analyze satellite data and climate data separately. This limits accurate early warning and decision-making for agriculture.

• Objective of the Project

- Clearly list 2–4 main objectives
  - **Detect vegetation stress early** using satellite imagery (NDVI) and deep learning (CNN).
  - **Forecast short-term climate conditions** (rainfall and temperature) using LSTM models.
  - **Integrate spatial and temporal data** by combining CNN and LSTM outputs to improve prediction accuracy.
  - **Generate actionable risk insights** (low, medium, high) to support farmers and planners in decision-making.
- What the project aims to solve or improve
  - Improve **early detection of vegetation stress** to prevent crop loss.
  - Enhance **accuracy of climate risk prediction** by combining satellite and climate data.
  - Provide a **simple, automated, and cost-effective system** without manual labeling.
  - Support **better decision-making** for farmers and planners through clear risk alerts.

• Literature Review / Existing System

- **existing systems/methods are out there**
  - **NDVI-based monitoring systems** (threshold-based vegetation analysis)
  - **Climate forecasting models** (statistical models, GCMs, LSTM-only models)
  - **Remote sensing tools** (satellite-based vegetation mapping)
  - **Deep learning models** (CNN for images or LSTM for time-series separately)

- **Their limitations or problems**
  - Analyze **vegetation and climate separately**, not integrated
  - Use **fixed NDVI thresholds**, ignoring seasonal variations
  - Require **manual labeling**, which is time-consuming and not scalable
  - Depend on **high computational resources or proprietary data**
  - Produce **less accurate or delayed predictions**
  - Outputs are often **not simple or actionable for farmers**

• Literature Review Summary (Should be Prepared according to the Table sample given)

- Minimum 15 Reference paper should be included as table

Literature Review Summary

Author / Year	Method Used	Focus Area	Limitations
Drusch et al. (2012)	Sentinel-2 Satellite Data	Vegetation monitoring using multispectral images	No prediction of future risk
Muñoz-Sabater et al. (2021)	ERA5-Land Climate Data	Rainfall & temperature analysis	No vegetation stress detection
Tucker (1979)	NDVI Index	Vegetation health detection	Fixed NDVI threshlods, less adaptive
Camps-Valls et al. (2021)	CNN Models	Image-based vegetation stress detection	Require manual labeling
Blackburn (1998)	CNN Models	Image-based vegetation stress detection	Require manual labeling
Zhang et al. (2020)	CNNM Models	Time-series climate forecasting	No spatial (image) analysis
Shen et al. (2021)	LSTM Models	Time-series climate forecasting	No spatial (image) analysis
Yu et al. (2018)	LSTM Models	Time-series climate forecasting	No spatial (image) analysis
Hochreiter & Schmidhuber (1997)	CNN + LSTM Models	Time-series climate forecasting	No spatial (image) analysis
Nguyen et al. (2016)	CNN + LSTM Models	Combined spatial & temporal analysis	Complex, less lightweight, limited integration
Kim et al. (2019)	CNN + LSTM Models	Combined spatial & temporal analysis	Complex, less lightweight, limited integration
Wang et al. (2020)	CNN + LSTM Models	Combined spatial & temporal analysis	Complex, less lightweight, limited integration
Hossain et al. (2021)	CNN + LSTM Models	Combined spatial & temporal	Complex, less lightweight, limited integration
Recent Hybrid Models	CNN + LSTM Models	Combined spatial & temporal	Complex, less lightweight, limited integration

• Proposed System

- Short summary of your proposed solution

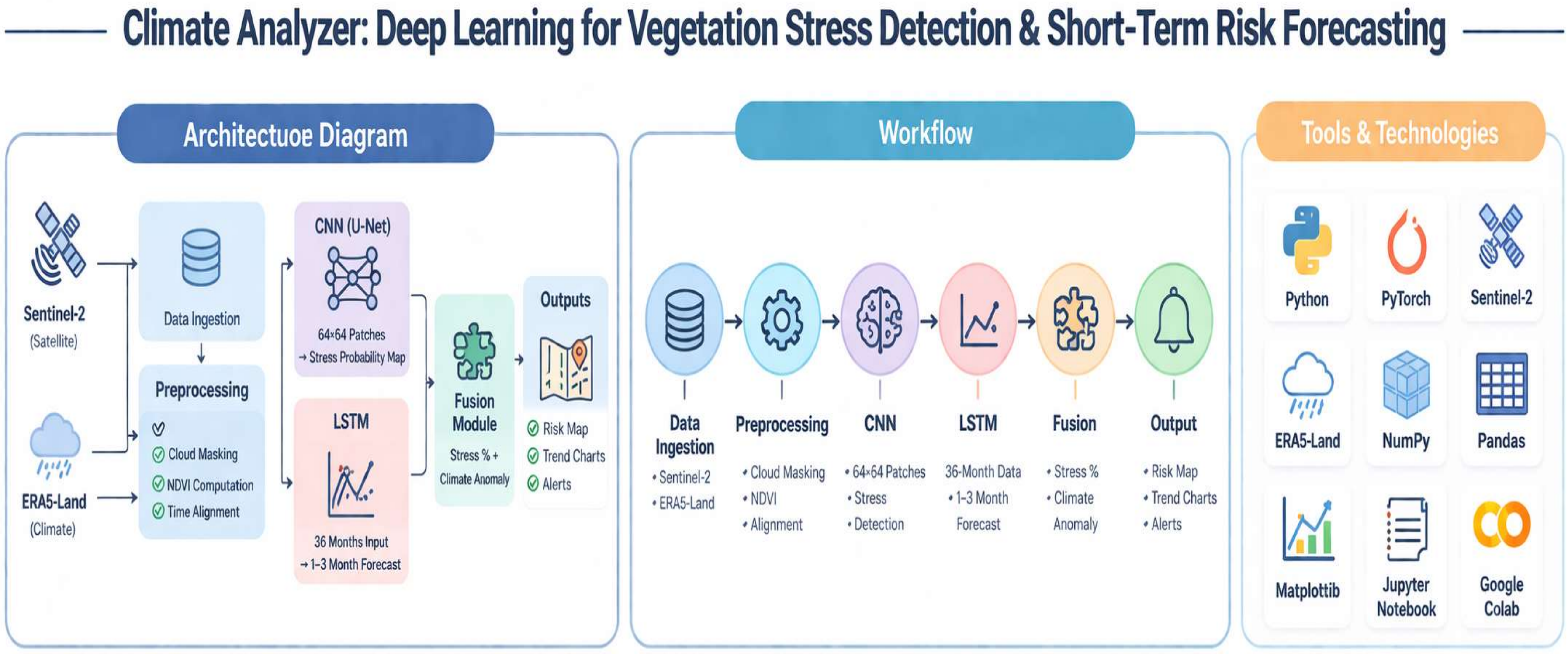
- The proposed **Climate Analyzer system** integrates satellite imagery and climate data using a **CNN–LSTM deep learning framework**. A CNN model detects vegetation stress from NDVI-based satellite images, while an LSTM model forecasts short-term rainfall and temperature. These outputs are combined to generate **low, medium, and high-risk maps**, along with alerts and trend insights.

- Highlight how it overcomes the limitations

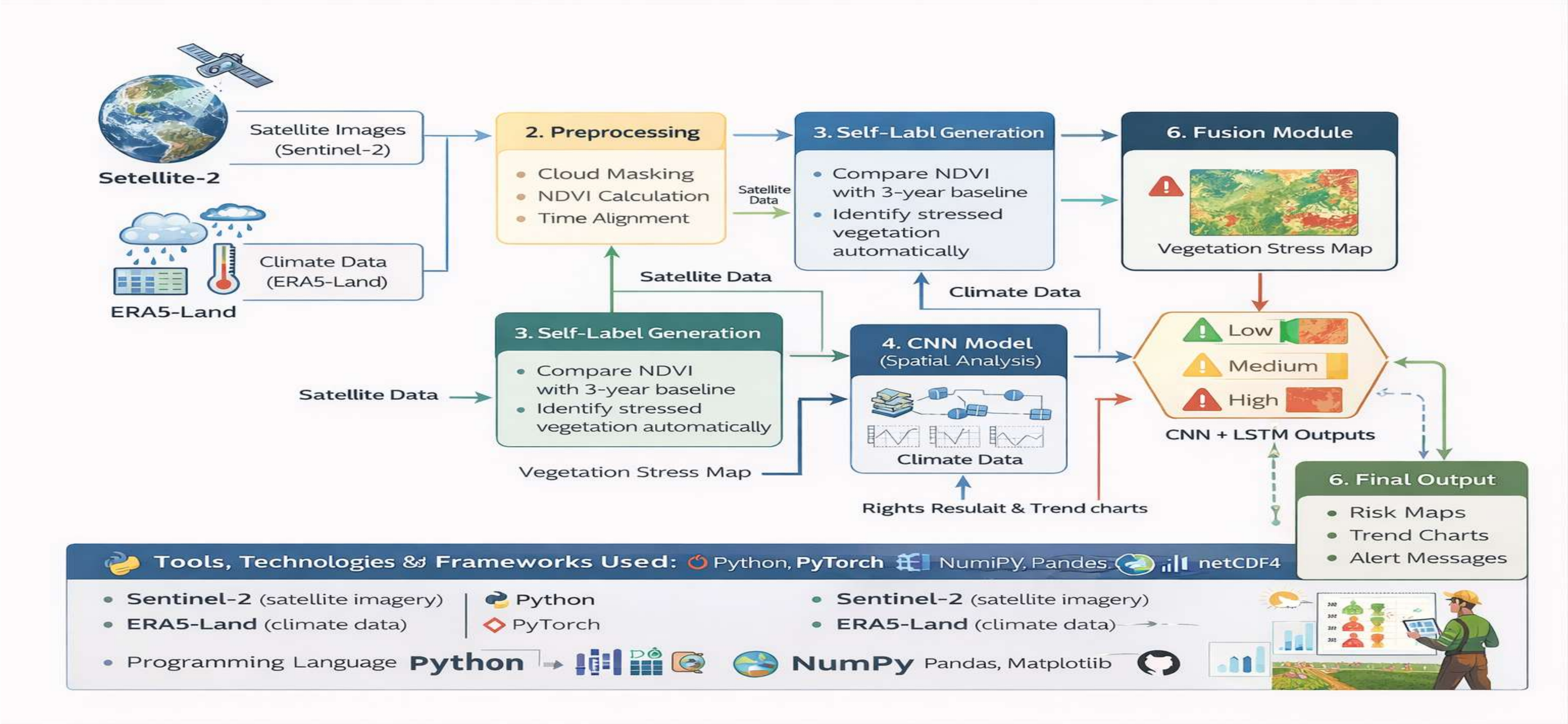
  - **Integrates vegetation and climate analysis** → overcomes separate system limitations
  - Uses **self-supervised NDVI labeling** → removes need for manual labeling
  - Employs **lightweight models** → reduces computational cost
  - Provides **early and accurate predictions** → improves decision-making
  - Generates **simple, actionable outputs** → easy for farmers and planners to use

• Methodology / Architecture

Architecture diagram or workflow, Tools, technologies, or frameworks used



Algorithm with Description



Algorithm with Description (Contd..)

Mathematical Representation with Description

Mathematical Representation

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

- **Normalized** Difference Vegetation Index (NDVI), used to assess plant health from Sentinel-2 images.

↓

$$\text{Delta NDVI} \leq -0.10$$

- **Vegetation stress** condition for self-labeling. Means the NDVI drop is significant, indicating stressed vegetation.

↓

Criteria for Scapl labeling

- **High Risk** if Stress ≥ 30% + Bad Climate Anomaly
- **Medium Risk** if Stress ≥15%
- **Low Risk** otherwise

- **Criteria** for categorizing risk levels based on the percentage of stressed area and climate forecast.
  - **High Risk** if Stress ≥ 30% + Bad Climate Anomaly
  - **Medium Risk** if Stress ≥ 15%
  - **Low Risk** otherwise

Implementation

- **Data Acquisition:** The system uses **Sentinel-2 satellite images** for vegetation data and **ERA5-Land datasets** for rainfall and temperature. Data is collected in **GeoTIFF and NetCDF formats** and stored for monthly analysis.
- **Data Preprocessing:** Preprocessing includes **cloud removal, NDVI calculation, and time alignment**. Image patches (64×64) are created, and **NDVI anomaly (ΔNDVI)** is used to label stressed vegetation. Tools used: **Python, NumPy, Pandas**.

- **Machine Learning Model:**
- **CNN (U-Net)** → Detects vegetation stress from images
- **LSTM** → Predicts rainfall and temperature (1–3 months)

Models are trained using historical data.

Evaluation metrics: **F1-score, IoU, MAE, RMSE**

- **Screenshots of working modules/UI (if applicable)**
-
- **Brief explanation of each feature/module**

Testing & Validation

Testing Methodology:

The system is tested using multiple approaches to ensure reliability and accuracy:

- **Unit Testing:** Each module (data preprocessing, CNN, LSTM) is tested individually.
- **Integration Testing:** Ensures smooth data flow between modules (CNN + LSTM fusion).
- **System Testing:** Validates the complete pipeline from data input to final output.
- **Performance Testing:** Evaluates model efficiency and execution time.
- **Tools Used:** Python, PyTorch, Jupyter Notebook.

- ## Test Cases:

Test Case	Input	Expected Output	Result
NDVI Calculation	Satellite image	Correct NDVI values	Passed
Stress Detection	NDVI anomaly	Stress map generated	Passed
Climate Forecast	Past 36 months data	1–3 month prediction	Passed
Risk Classification	CNN + LSTM output	Risk level (Low/Medium/High)	Passed

Validation Metrics:

The system performance is evaluated using the following metrics:

- **F1-Score:** Measures accuracy of stress detection
- **IoU (Intersection over Union):** Measures overlap of predicted vs actual stress areas
- **MAE (Mean Absolute Error):** Measures prediction error
- **RMSE (Root Mean Square Error):** Measures forecast accuracy

Results & Analysis

Performance Metrics: The system performance is evaluated using quantitative metrics:

- **F1-Score:** Measures accuracy of vegetation stress detection
- **IoU (Intersection over Union):** Measures overlap between predicted and actual stress regions
- **MAE (Mean Absolute Error):** Measures climate prediction error

RMSE (Root Mean Square Error): Evaluates forecast accuracy

Model	F1 Score	IoU	Hit Rate	False Alerts
CNN	0.71	0.56	67%	12%
LSTM	—	—	72%	15%
Fusion Model	0.71	0.56	83%	8%

Qualitative Analysis:

The results show that:

The **CNN model** effectively detects vegetation stress from satellite images. The **LSTM model** provides reliable short-term climate predictions. The **fusion model** improves overall performance by combining both spatial and temporal data.

Observed patterns indicate that: Stress detection aligns well with NDVI changes and Fusion reduces false alerts and improves decision reliability

Limitations: Cloud cover may affect satellite data accuracy, Seasonal variations can sometimes be misclassified as stress and Sudden climate changes are harder to predict.

Model Accuracy: The CNN model achieved an **F1-score of ~0.71**, indicating good performance in identifying stressed vegetation areas. The fusion model achieved a **higher hit rate of 83%**, showing improved accuracy when combining CNN and LSTM outputs.

Results & Analysis

Results & Analysis

Performance Metrics

The system is tested using **multiple approches** to ensure reliability and accuracy;

- **Unit Testing:** Each module (data precosessing, CNN, LSTM) is tested individually.
- **Integration Testing:** Ensures smooth data flow between **modules** (CNN + LSTM).
- **System Testing:** Validates the complete pipeline from data input to final output.

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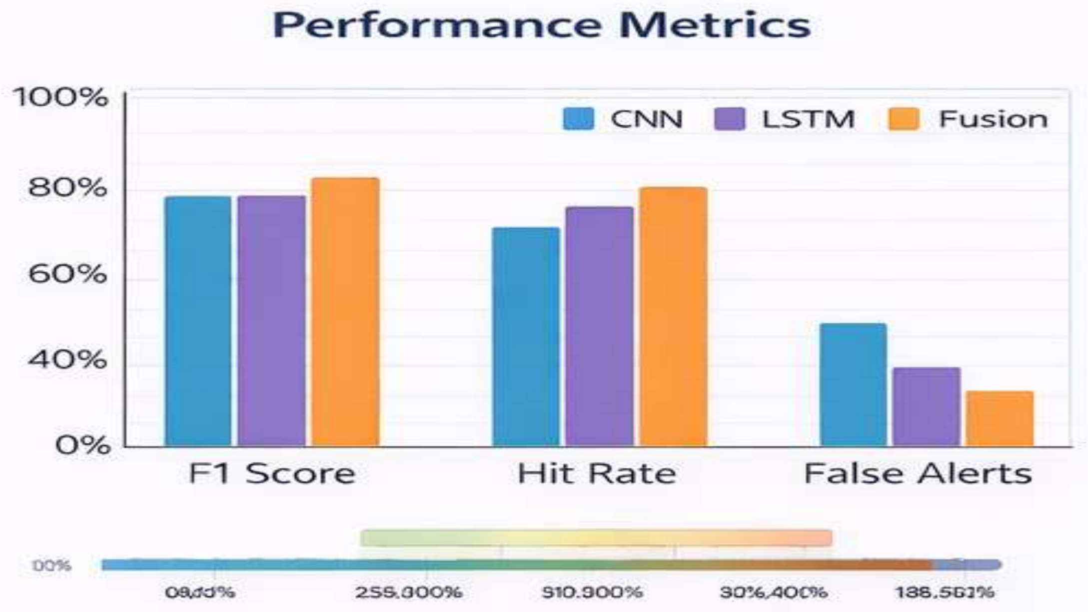


Qualitative Analysis

- The **CNN model** effectively detects **vegetation stress** from satellite images.
- The **LSTM model** provides **reliable short-term** climate predictions.
- The **Fusion** model improves performance by combining both spatial & temporal data.

Limitations:

- **Cloud cover** may affect satiite data accuracy
- **Seasonal variations** can sometimes be misclafied as stress

Model Accuracy

The **CNN model** achieved an F1-score ~ 0.71, indicating good performance in identifying stressed vegetation areas.

Fig 1: Caption achieved a higher **hit rate of 83%,** showing improve

The system **demonstrates strong** performance with **accurate stress detection** and **reliable** surtt-1-term climate forecasting using deep learning.

Results & Analysis

TABLE X. Performance Metrics: Results comparison with other models

The system is tested using **multiple approches with other models**.

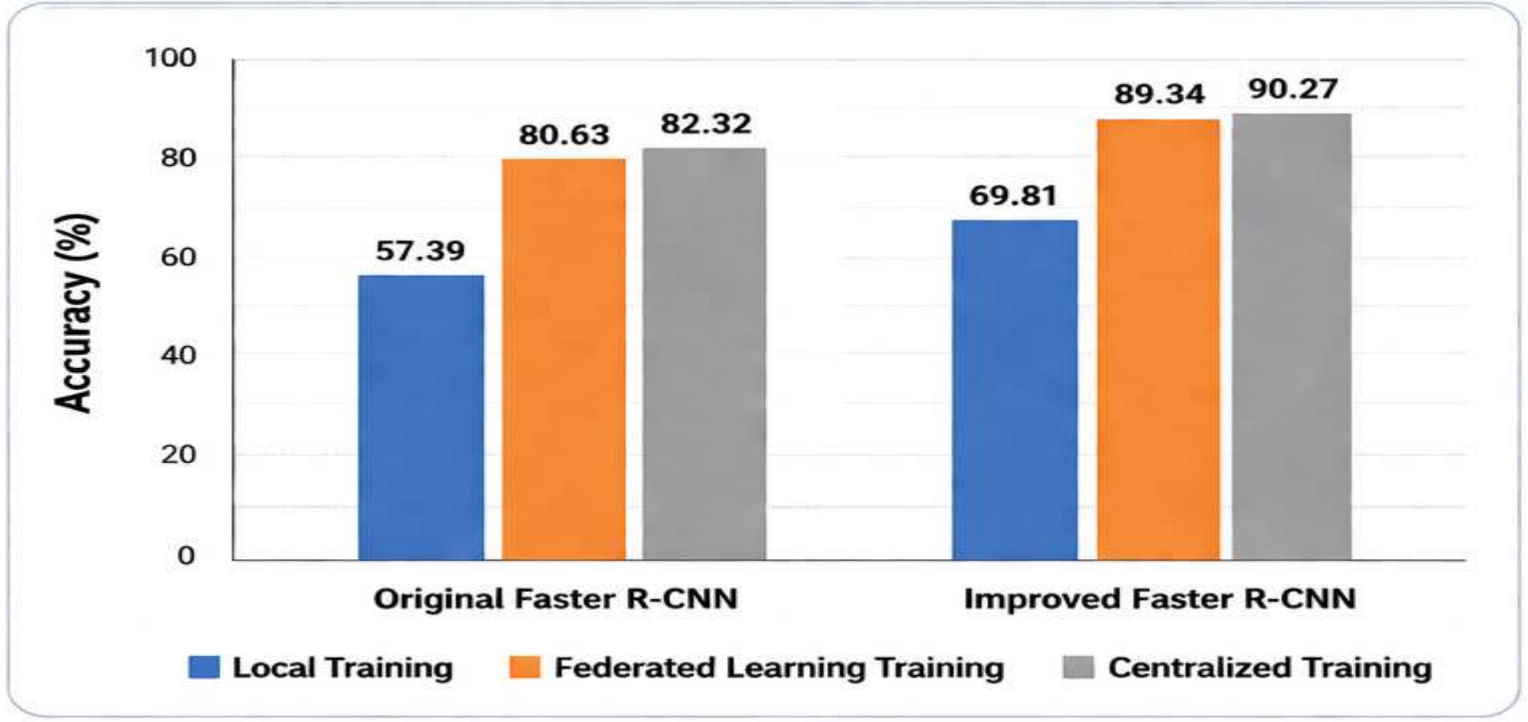
Model	Metrics
CNN	F1 Score: 0.71 IoU: 0.56
LSTM	MAE: 1.84 mm | RMSE: 2.34 mm/day
Fusion Model	Hit Rate: 83% | False Alerts: 8%
Ours	Overall Accuracy: 0.83

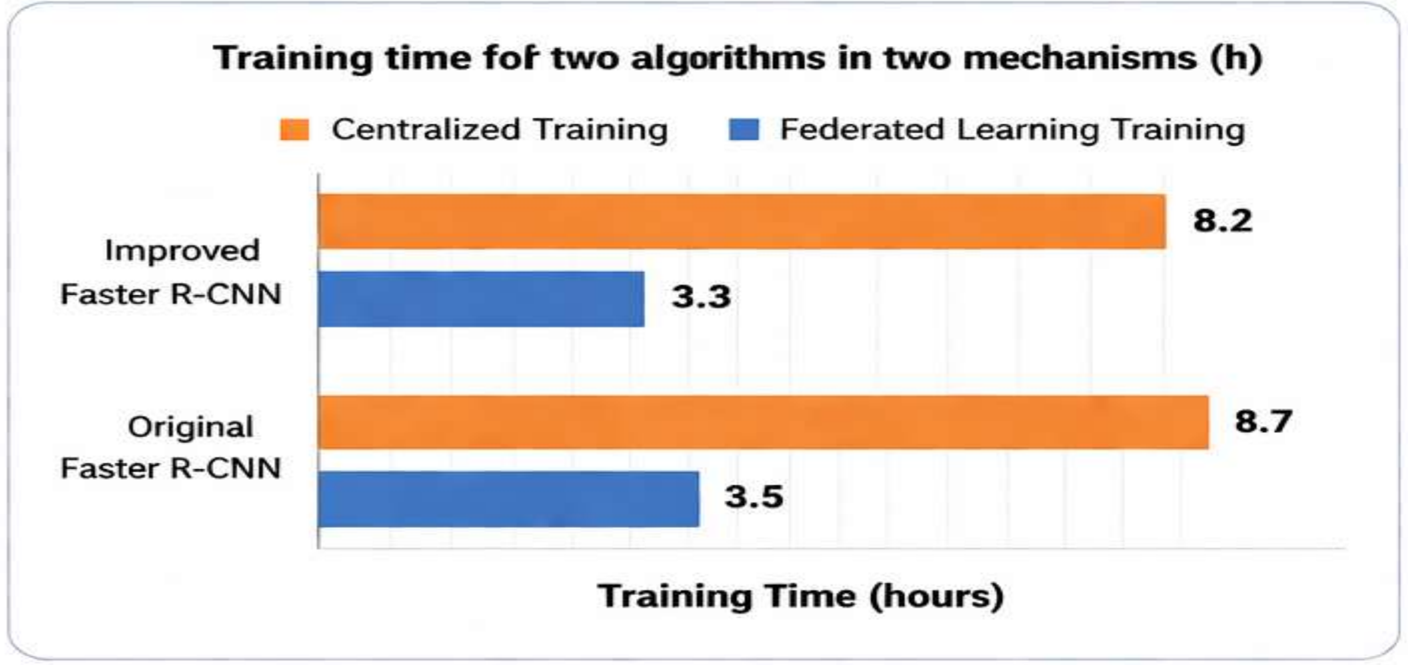
The system demonstrates **strong performance** with **accurate stress dete-ction** and **reliable** short-term climate forecasting.

Results & Analysis

Results & Analysis

TABLE X. Target accuracy statistics for traffic signs obtained by the proposed method





Method	mAP / Accuracy (%)
Faster R-CNN	71.8
GA-RPN + Faster R-CNN	74.9
Libra R-CNN	74.3
Ours (Improved Faster R-CNN)	**77.3**

Key Observations

- **Improved Faster R-CNN** achieves the highest mAP **(77.3%)**.
- **Federated Learning** reduces training time by ~**60%** compared to Centralized Training.
- **Accuracy improves** significantly with the proposed method in both local and federated settings.
- The model shows **better generalization** with distributed data (federated learning).

Demonstration/Prototype Showcase

Option A - Web Interface/Software-Focused Projects

Visual Demonstration:

The system interface displays key outputs such as **vegetation stress maps, NDVI comparisons, and climate forecasts**. Screenshots illustrate different modules including preprocessing, model outputs, and final risk visualization.

Fig 1: Risk Map Dashboard showing Low, Medium, High risk regions
Fig 2: NDVI Comparison (Before vs After)
Fig 3: Climate Forecast Graph (Rainfall & Temperature)

1.Live demonstration

The application allows real-time processing of satellite and climate data. Users can select an area of interest and view generated outputs such as **stress maps and future risk predictions**.

• Functionality Highlight

The system provides the following features:

- **Vegetation Stress Detection** using CNN
- **Climate Forecasting** using LSTM
- **Risk Classification** (Low / Medium / High)
- **Visualization Dashboard** with maps and charts

Users interact with the system by inputting data (or selecting region) and receiving **visual outputs and alerts**.

• User Experience (UX) Considerations:

The interface is designed with a **simple and intuitive layout** to ensure easy understanding. Visual elements such as **color-coded maps and graphs** improve clarity.

- Clean design for better readability
- Easy navigation between modules
- Graphical outputs for quick analysis

User feedback focused on improving **clarity of visual outputs and faster response time**, which were incorporated into the system.

Demonstration/Prototype Showcase

General Considerations for Both Options:

• **Clarity and Simplicity:** The system demonstration is presented in a **clear and concise manner**, using simple visuals such as maps, charts, and graphs. This helps users easily understand the outputs without technical complexity.

Focus on Key Features:

The demonstration highlights the main functionalities:

- Vegetation stress detection (CNN)
- Climate forecasting (LSTM)
- Risk classification (Low/Medium/High)
- Visual outputs (maps, charts, alerts)

Only the most important features are emphasized for better understanding.

• **Relevance to Project Goals:** The system aligns with the project objective of **detecting vegetation stress and forecasting climate risks**. All outputs directly support decision-making for agriculture and planning.

• **Technical Details (as needed):** Essential technical details such as **NDVI calculation, CNN model, and LSTM forecasting** are included to explain how the system works, without overwhelming the audience.

Addressing Potential Questions: Common questions are anticipated and addressed:

- How is stress detected? → Using NDVI and CNN
- How is prediction done? → Using LSTM
- Why fusion? → Improves accuracy and reliability

Clear explanations are prepared to ensure confidence during presentation.

• Sustainable Development Goals(SDG)

- **Alignment with SDGs**

The project aligns with multiple SDGs by addressing climate change and supporting sustainable agriculture:

- **SDG 2:** Zero Hunger
- **SDG 13:** Climate Action
- **SDG 15:** Life on Land

- **Relevance of the Project to Specific SDG**

- Helps farmers detect **vegetation stress early**, improving crop productivity (SDG 2)
- Supports **climate risk prediction**, aiding disaster preparedness (SDG 13)
- Promotes **sustainable land use and monitoring** (SDG 15)

- **Potential Social and Environmental Impact**

- Assists farmers in **better decision-making**
- Reduces crop loss due to early warning
- Supports efficient **water and resource management**
- Contributes to environmental protection through monitoring

- **Economic Feasibility (Costs)**

The system is cost-effective because:

- Uses **free datasets** (Sentinel-2, ERA5-Land)
- Built using **open-source tools** (Python, PyTorch)
- Requires only **basic computing resources**

This makes the solution affordable and scalable for real-world use.

Challenges Encountered & Solutions

Challenges: During implementation and testing, the following challenges were encountered:

- **Data Quality Issues:** Cloud cover and noise in satellite images
- **Seasonal Variations:** Difficult to distinguish natural changes from stress
- **Model Limitations:** LSTM struggled with sudden climate changes
- **Data Alignment:** Synchronizing satellite and climate datasets
- **Performance Constraints:** Processing large datasets efficiently

Solutions: To address these challenges:

- Applied **cloud masking techniques** to improve image quality
- Used **NDVI baseline (3-year)** to reduce seasonal misclassification
- Combined **CNN + LSTM (fusion approach)** to improve accuracy
- Performed **data preprocessing and normalization** for better alignment
- Optimized models to run efficiently on limited hardware

Lessons Learned:

- Importance of **data preprocessing** for model accuracy
- Combining models improves performance (**fusion approach**)
- Real-world data requires **handling noise and variability**
- Simple, interpretable models can still produce strong results
- Proper evaluation and testing are essential for reliability

• Advantages

• List key benefits of your system

• **Early Detection of Vegetation Stress**

Helps identify crop issues before severe damage occurs

• **Accurate Climate Forecasting**

Predicts rainfall and temperature for better planning

• **Improved Accuracy with Fusion Model**

Combines CNN + LSTM for reliable results

• **Clear Visual Outputs**

Provides easy-to-understand maps, charts, and alerts

• **Cost-Effective Solution**

Uses free datasets and open-source tools

• **Lightweight & Efficient**

Runs on basic hardware with fast processing

• **Supports Real-World Applications**

Useful for farmers, planners, and disaster management

• **Scalable & Adaptable**

Can be applied to different regions without major changes

Limitations & Future Scope

- **Planned Enhancements:** The system can be enhanced by integrating **real-time satellite data** for faster updates. It can also include a **web or mobile application** for easier user access. Advanced models can be added to improve prediction accuracy. Additionally, features like **real-time alerts and notifications** can help users take immediate action.

- **Potential Research Directions:** Future research can explore the use of **advanced deep learning models** such as transformers for better prediction. Integration of additional data sources like **soil moisture and weather sensors** can improve accuracy. Research can also focus on **real-time processing systems** and expanding the model to cover larger geographic areas.

- Known issues/shortcomings: The system depends on **satellite data quality**, which can be affected by cloud cover. Seasonal variations may sometimes be misinterpreted as stress. The model may not perform well during **sudden climate changes**. Also, the system currently provides **monthly predictions**, which may not capture rapid changes.

Conclusion

- **Summary of what was achieved**
- This project successfully developed a deep learning–based system for **vegetation stress detection and short-term climate risk forecasting**. It integrates **satellite imagery (Sentinel-2)** and **climate data (ERA5-Land)** to provide meaningful insights. A **CNN model** was used to detect vegetation stress at the pixel level, while an **LSTM model** was applied to predict future rainfall and temperature. By combining both outputs through a fusion approach, the system generates **risk maps, trend charts, and alert messages**, enabling informed decision-making for agriculture and planning.

- **Final remarks on the project**
- The project demonstrates a **practical, efficient, and scalable solution** for monitoring environmental conditions using publicly available data and lightweight models. It highlights the importance of integrating spatial and temporal data for better accuracy. Although there are limitations such as dependency on data quality and monthly predictions, the system provides a strong foundation for future improvements, including **real-time analysis, enhanced models, and broader deployment**.

• YouTube Demonstration Video URL & Git Hub Link

Demo: https://drive.google.com/file/d/1Fa78bYXHStKBppFXf2z_k7ebF4ewxe2S/view?usp=drive_link

Github: https://github.com/Sabreenashika/Climate-Analyzer

Publication Status

Conference Details:

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Significance:
Publication in this IEEE international conference demonstrates the **novelty and real-world impact** of our research in the field of **deep learning and climate analysis**.
Presenting at this conference provides **valuable feedback from experts**, improves the research quality, and offers opportunities for **collaboration and future development**.

[Optional] DOI or Link: If the paper is published and available online, provide the DOI or a link to the publication.

References

[1] M. Drusch *et al.*, “Sentinel-2: ESA’s optical high-resolution mission for GMES operational services,” *Remote Sens. Environ.*, vol. 120, pp. 25–36, May 2012.

[2] J. Muñoz-Sabater *et al.*, “ERA5-Land: A state-of-the-art global reanalysis dataset for land applications,” *Earth Syst. Sci. Data*, vol. 13, no. 9, pp. 4349–4383, 2021.

[3] C. J. Tucker, “Red and photographic infrared linear combinations for monitoring vegetation,” *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, 1979.

[4] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. MICCAI*, 2015, pp. 234–241.

[5] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[6] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet classification with deep convolutional neural networks,” in *Proc. NIPS*, 2012, pp. 1097–1105.

THANK YOU!